

Math Bootcamp: Content to be Covered

For Incoming MSc Economics Students

Math Camp 2026

Overview

- **Objective:** Bridge the gap between a weaker undergraduate math background and MSc-level economics courses.
- **Duration:** 3 hours per day for 5 days.
- **Method:** Focus on intuition, fundamentals, and confidence-building. Rigorous proofs and advanced techniques are deferred to fall courses; only simple proofs are included where necessary.
- **Emphasis:** Conceptual topics to be covered throughout graduate microeconomics, macroeconomics, econometrics, and mathematical economics.

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1 Day 1 — Foundations, Functions, and Calculus I

Approximate duration: 3 hours.

1.1 Notation and Number Systems

- Common sets: $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{R}^n$.
- Symbols: $\in$, $\subset$, $\forall$, $\exists$, $\Rightarrow$, $\Leftrightarrow$.
- Vectors in $\mathbb{R}^n$: notation $\mathbf{x} = (x_1, \dots, x_n)$; scalars versus vectors.

1.2 Logic and Mathematical Statements

- Necessary versus sufficient conditions; “only if” and “if.”
- Contrapositive: $\neg Q \Rightarrow \neg P$ is equivalent to $P \Rightarrow Q$ (conceptual).
- “If and only if” ($\Leftrightarrow$) and equivalence of statements.
- How economists use these terms (e.g., first-order conditions necessary for an interior optimum; sufficiency under concavity).

1.3 Sets and Set Operations

- Sets, elements, subsets, and equality of sets.
- Union, intersection, complement, and difference.
- Cartesian product $A \times B$ (preview of product spaces on Day 3).
- Economic examples: consumption sets, production sets, budget sets as sets in $\mathbb{R}^n$.

1.4 Functions: Definitions and Properties

- Function as a rule: domain, codomain, and range.
- Injective (one-to-one), surjective (onto), and bijective functions.
- Operations on functions: sum, scalar multiple, product, and composition.
- Inverse functions when a map is bijective.

1.5 Supremum and Infimum

- Maximum/minimum versus supremum (sup) and infimum (inf).
- Least upper bound and greatest lower bound (intuitive definitions).
- When a supremum is attained versus only approached.
- Economic relevance: bounds on feasible choices, value functions, and equilibrium objects defined as suprema.

1.6 Homogeneous Functions of Degree k

- Definition: $f(t\mathbf{x}) = t^k f(\mathbf{x})$ for $t > 0$.
- Checking homogeneity (degree zero, one, and general k).
- Euler’s theorem for C^1 homogeneous functions: $\sum_i x_i \frac{\partial f}{\partial x_i} = k f(\mathbf{x})$ (statement).

- **Economic links:** constant returns to scale (production homogeneous of degree 1); homogeneous preferences; Cobb–Douglas as degree 1 in total expenditure form; marginal products and scaling of inputs; homogeneity in estimable demand systems and index numbers (high-level).

1.7 Algebra Review and Diagnostic

- Self-assessment of algebra and calculus readiness.
- Logarithms and exponentials: definitions, rules, and economic relevance.
- Basic function types (polynomial, power, log, exponential).
- Simple univariate optimization as a preview.

1.8 Limits and Continuity (Introductory)

- Intuitive notion of a limit.
- Continuity of a function at a point and on an interval.
- Why continuity matters for economic models (existence, stability of solutions).
- One-sided limits and basic limit laws (conceptual, not formal ε - δ).

1.9 Derivatives: Intuition and Rules

- Derivative as instantaneous rate of change and slope of tangent.
- Differentiability versus continuity.
- Core differentiation rules: power, product, quotient, and chain rules.
- Derivatives of composite economic functions.
- Higher-order derivatives and economic interpretation (marginal versus curvature).

1.10 Partial Derivatives

- Functions of several variables.
- Level curves / contour map: $\mathcal{L}_c = \{(x, y) : f(x, y) = c\}$; 3D picture as a horizontal slice of $z = f(x, y)$.
- Partial derivative as marginal effect holding other variables fixed.
- Cross-partial derivatives and symmetry (Schwarz's theorem, stated not proved).
- Economic interpretation: marginal products, marginal utilities, marginal rates.

1.11 Total Differential

- Setup: $\psi(t) = f(x + t dx, y + t dy)$; actual change $\Delta f = \psi(1) - \psi(0)$.
- Chain rule at $t = 0$: $\psi'(0) = f_x dx + f_y dy =: df$; gradient $\nabla f = (f_x, f_y)$.
- Approximation: $\psi(1) - \psi(0) \approx df$ for small dx, dy .
- Simple $\psi(t)$ graph: Δf vs. linear approx. $\psi'(0)$.
- Application: $dY = MP_K dK + MP_L dL$ (Cobb–Douglas numeric example); later uses include dU and budget differentials.

1.12 Gradient Geometry

- Contour-map picture: ∇f at P on $\mathcal{L}_c$, tangent to level curve.
- Derivation ($\perp$): if $\mathbf{r}(t) \in \mathcal{L}_c$, then $0 = \frac{d}{dt}f(\mathbf{r}(t)) = \nabla f \cdot \mathbf{r}'(t)$.
- Derivation (uphill): $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = \|\nabla f\| \cos \theta$; maximum when $\mathbf{u} \parallel \nabla f$.
- Along a level curve, $D_{\mathbf{u}}f = 0$; MRS as slope of indifference curve.

1.13 Directional Derivatives

- Rate of change of f along a chosen unit direction $\mathbf{u}$ (rate version of df with $(dx, dy) = t\mathbf{u}$).
- Definition as a limit along the line $(a, b) + t\mathbf{u}$.
- Derivation: same chain rule as df ; $\varphi(t) = f(a + tu_1, b + tu_2) \Rightarrow D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$.

2 Day 2 — Calculus II and Linear Algebra: Implicit Functions, Integration, and Matrix Methods

Approximate duration: 3 hours.

2.1 Implicit Function Theorem

- Explicit versus implicit definitions of relationships.
- What problem the Implicit Function Theorem solves: local solvability and comparative statics.
- Statement of the theorem in one and several variables (informal).
- Sign of derivative or Jacobian and direction of local response.
- Connection to economic comparative statics (Slutsky-type decompositions, IS–LM shifts).

2.2 Integration

- Riemann integral: area under a curve.
- Fundamental Theorem of Calculus (statement and intuition).
- Definite versus indefinite integrals.
- Integration as accumulation.
- Link to probability: PDF and CDF; CDF as integral of PDF; expected value as integral (conceptual).
- Inverse relationship between differentiation and integration (conceptual).

2.3 Leibniz Rule and Interchanging Differentiation and Integration

- Leibniz product rule (review): derivative of a product of functions.
- **Fixed limits:** differentiation under the integral sign when the integrand depends on a parameter but the bounds do not.
- **Variable limits:** when the upper and lower bounds of integration depend on the choice variable (parameter), the derivative has two extra *endpoint* terms from the moving limits, plus an integral of the partial derivative of the integrand.

- Informal statement: if $F(\theta) = \int_{a(\theta)}^{b(\theta)} f(x, \theta) dx$, then $F'(\theta)$ equals $f(b(\theta), \theta) b'(\theta) - f(a(\theta), \theta) a'(\theta)$ plus $\int_{a(\theta)}^{b(\theta)} \frac{\partial f}{\partial \theta}(x, \theta) dx$ (under regularity conditions).
- Economic motivation: consumer surplus, areas under shifting demand curves, and other objects where both the integrand and the integration region change with a parameter.
- When we may interchange $\frac{d}{d\theta}$ and $\int$: continuity, differentiability, and boundedness (conceptual); dominated convergence as the measure-theoretic idea behind safe interchange (stated, not proved).
- When interchange fails: discontinuities, unbounded integrands, or limits that move irregularly.
- Distinction from Fubini (swap order of integrals), MCT, and LDCT (swap limit and integral); preview on Day 3 via uniform convergence.

2.4 Mean Value Theorem

- Geometric intuition: existence of a tangent parallel to a secant.
- Rolle's theorem as a special case (statement only).
- Economic uses of the Mean Value Theorem: average versus marginal quantities; bounding changes over intervals.

2.5 Taylor Expansion and Linearization

- First-order Taylor approximation.
- Linearization of nonlinear functions around a point.
- Interpretation as local first-order comparative statics.
- When linearization is useful versus when it fails (curvature, nonlinearity).
- Preview of second-order Taylor expansion (conceptual only).

2.6 Vectors and Matrices

- Vectors in $\mathbb{R}^n$: notation, components, and geometric interpretation.
- Matrices: dimensions, entries, and matrix–vector multiplication.
- Basic matrix operations: addition, scalar multiplication, and multiplication.
- Transpose of a matrix or vector.
- Identity matrix and role in matrix algebra.

2.7 Linear Systems, Inverses, and Determinants

- Systems of linear equations in matrix form ($A\mathbf{x} = \mathbf{b}$).
- Inverse of a square matrix: definition and when it exists.
- Determinant: definition for 2×2 and 3×3 matrices; link to invertibility (conceptual).
- Linear independence and dependence of vectors (brief, intuitive treatment).
- Rank of a matrix (conceptual: number of linearly independent rows or columns).

2.8 Jacobians and Economic Applications of Linear Algebra

- Jacobian matrix as a collection of partial derivatives.
- Dot product and its economic interpretation (e.g., expenditure, inner products in econometrics).
- Quadratic forms (brief introduction: notation and positive/negative definiteness, stated not proved).
- How matrix notation simplifies multivariate comparative statics; return to the Implicit Function Theorem in matrix form.

3 Day 3 — Real Analysis: Topology, Convergence, and Existence

Approximate duration: 3 hours.

3.1 Metric Spaces and Basic Topology in $\mathbb{R}$ and $\mathbb{R}^2$

- Points, distances, and neighborhoods.
- Open sets: definition and intuition.
- Closed sets: definition and intuition.
- Interior, boundary, and closure (conceptual).
- Economic relevance: budget sets, feasible sets, production sets; open versus closed constraints.

3.2 Convergence

- Sequences and limits in $\mathbb{R}^n$.
- Pointwise convergence of sequences of functions.
- Uniform convergence: definition and intuition.
- Why uniform convergence matters: exchange of limits; preservation of continuity; interchange of limit and integral; interchange of differentiation and integration (stated, not proved).

3.3 Continuity (Rigorous Intuition)

- ε - δ definition (light treatment; primarily conceptual).
- Continuity on sets versus at a point.
- Continuity of demand, supply, profit, and value functions.
- Discontinuities: types and economic meaning.
- When discontinuities create problems for optimization or equilibrium.

3.4 Compactness

- Definition of compact set (in $\mathbb{R}^n$, via closed and bounded for this level).
- Heine–Borel property (statement).
- Why compactness matters: existence of maxima and minima for continuous functions.

- Weierstrass Extreme Value Theorem (statement).
- Maximum and minimum of continuous functions on compact domains.

3.5 Product Spaces

- Cartesian products $\mathbb{R}^n$.
- Notation for vectors and product domains.
- Constraints and choice sets as subsets of product spaces.
- Separability across dimensions (conceptual).

3.6 Separation Theorem

- Convex sets in $\mathbb{R}^n$ (definition sufficient for separation; detailed convexity on Day 4).
- Hyperplane separation of disjoint convex sets (picture-based intuition).
- Supporting hyperplanes.
- Application to welfare economics: Pareto efficiency and competitive equilibrium.
- First and Second Welfare Theorems (high-level).

3.7 Fixed Point Theorems

- Fixed point: definition and economic motivation (equilibrium as fixed point).
- At least one fixed point theorem (e.g., Brouwer or contraction mapping).
- Role in proving existence of equilibrium.
- Contraction mapping as a special case (if covered).

4 Day 4 — Optimization

Approximate duration: 3 hours.

4.1 Convex Sets and Convex Functions

- Definition and geometry of convex sets.
- Convex and concave functions.
- Upper and lower contour sets (level sets).
- Why convexity or concavity guarantees global optima from first-order conditions.

4.2 Quasiconcavity and Quasiconvexity

- Definition of quasiconcave and quasiconvex functions via contour sets.
- Relationship between concavity/convexity and quasiconcavity/quasiconvexity.
- Why quasiconcavity is the standard regularity condition in utility maximization.
- When first-order conditions are sufficient for a global optimum under quasiconcavity.
- Strict quasiconcavity and uniqueness of optima (conceptual).

- Invariance under monotone transformations.
- Distinction between quasiconcavity of the objective and convexity of the feasible set in constrained problems.

4.3 Unconstrained Optimization

- Local versus global optima.
- First-order conditions: necessary conditions for interior optima.
- Second-order conditions: curvature and nature of critical points.
- Concavity and convexity of objective functions and uniqueness of optima.
- Economic optimization problems as a class (profit, utility, cost minimization).

4.4 Constrained Optimization: Equality Constraints

- Feasible set defined by equality constraints.
- Lagrange multiplier method: setup and interpretation.
- Lagrange multipliers as shadow prices or marginal values of constraints.
- Necessary conditions for constrained optima (Lagrange theorem, stated).
- Utility maximization subject to budget constraint (as problem type).

4.5 Constrained Optimization: Inequality Constraints

- Feasible set with inequalities.
- Binding versus non-binding constraints.
- Complementary slackness (conceptual).
- Karush–Kuhn–Tucker (KKT) conditions (statement and interpretation).
- Nonnegativity constraints and corner solutions.

4.6 Envelope Theorem

- **Placement:** after unconstrained and constrained optimization (FOCs, Lagrange, KKT).
- Value function of a parameterized problem: $V(\theta) = \max_x f(x; \theta)$ subject to constraints.
- **Unconstrained envelope theorem:** $\frac{dV}{d\theta} = \frac{\partial f}{\partial \theta} \Big|_{x=x^*(\theta)}$ (no need to differentiate the optimizer $x^*(\theta)$ directly).
- **Constrained envelope theorem:** $\frac{dV}{d\theta} = \frac{\partial \mathcal{L}}{\partial \theta} \Big|_{(x^*, \lambda^*)}$ with binding multipliers capturing constraint movements.
- Interpretation: shadow prices and comparative statics from the optimized value.
- Economic examples: profit as a function of a cost shifter or output price; indirect utility as a function of income and prices; expenditure/cost functions and Shephard-type logic (conceptual).
- Contrast with naive chain rule through $x^*(\theta)$: envelope theorem drops terms that vanish at the optimum.

5 Day 5 — Dynamic Programming

Approximate duration: 3 hours.

5.1 Motivation and Framework

- Sequential decision-making over time.
- Intertemporal trade-offs.
- Finite versus infinite horizon (finite focus for bootcamp).
- Markov structure: current state summarizes relevant history.

5.2 State Variables versus Control Variables

- State: variables that evolve according to law of motion.
- Control: variables chosen each period.
- Transition equation or law of motion.
- Feasibility constraints in dynamic problems.
- Classic problem types: consumption–savings, capital accumulation, resource extraction.

5.3 Bellman Equation

- Value function: maximum attainable payoff from a given state.
- Principle of optimality (Bellman’s idea).
- Recursive formulation of dynamic problems.
- Bellman equation for finite-horizon problems.
- Terminal condition and backward solution logic.

5.4 Backward Induction

- Solving finite-horizon problems from the last period backward.
- Relationship between period-by-period optimization and Bellman recursion.
- Policy function: mapping from state to optimal control.

5.5 Value Function Iteration (Conceptual)

- Idea of iterating on the Bellman operator.
- Convergence to fixed point of Bellman equation (intuition only, no algorithm).
- Distinction between solving finite-horizon and infinite-horizon problems.

5.6 Two-Period Models as Building Blocks

- Structure of two-period consumption–savings problems.
- Euler equation (conceptual introduction if appropriate).
- Link between dynamic programming and standard intertemporal optimization.

6 Cross-Cutting Themes (All Days)

- Mathematical notation used in economics graduate courses.
- Reading and interpreting first-order conditions, second-order conditions, and comparative statics.
- Connecting mathematical objects to economic meaning (marginal, shadow price, value function).
- Confidence with standard problem templates used in fall courses.

7 Suggested Time Allocation

Day	Block 1 (~1 hr)	Block 2 (~1 hr)	Block 3 (~1 hr)
1	Notation; logic; sets; functions	Sup/inf; homogeneous functions; limits	Derivatives; partials; total differential; directional
2	Implicit Function Theorem; MVT	Integration; Leibniz rule; Taylor	Vectors; matrices; Jacobians
3	Open and closed sets	Convergence; continuity; product spaces	Compactness; separation; fixed points
4	Convexity; quasiconcavity	Unconstrained optimization	Lagrange; KKT; envelope theorem
5	Framework; state and control	Bellman equation	Backward induction; VFI idea